MILITARY SPECIFICATION

RESIN, POLYESTER, BISPHENOL - A TYPE

1. SCOPE

1.1 This specification covers the requirements of bisphenol -A derived polyester resins.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Specifications:

Federal

NN-P-515	Plywood, Container Grade
PPP-C-96	Cans, Metal, 28 Gage and Lighter
PPP-B-601	Boxes, Wood, Cleated - Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-D-760	Drums & Pails, Metal (5 and 16.64 Gallon)

Standards

Federal

Fed. Test Method Std. No. 406 - Plastics: Methods of Testing

Military

MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129, Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publication required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material - The material shall be a polyester resin based on a reacted bisphenol -A derivative.

3.2 Code number - To identify the chemical composition and manufacturing process of a resin submitted for acceptance, the manufacturer shall assign a code number to the resin. The code number may also include a trade name. Any changes in chemical composition or manufacturing process shall require the assignment of a new code number.

FSC 9330

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3.3 Chemical resistance - When applicable, suitability of the material for use in chemically corrosive atmospheres shall be as specified by the procuring activity in the procurement document or contract. Method of determining conformance to the requirements shall be as specified by the procuring activity.

3.4 Property requirements - Specimens prepared from sheets cast in accordance with the manufacturer's instructions shall exhibit the properties specified in Table I when tested in accordance with the appropriate tests of 4.6.

Table I - Property Requirements

Property	Requirement	Paragraph Reference to Applicable Test.
Flexural strength, psi, min.	14,000	4.6.3
Flexural modulus, psi, min.	0.40×10^6	4.6.3
Tensile strength, psi, min.	8,000	4.6.4
Elongation, %, min.	1.5	4.6.4
Deflection temperature under load, at 264 psi fiber stress, °F, min.	260	4.6.5
Water absorption, % max.	0.25	4.6.6

3.5 Instruction sheet - Unless otherwise specified, the manufacturer shall submit to the procuring activity two copies of an instruction sheet described below. The instruction sheet shall include pertinent instructions on the storage and handling of the material as well as complete information on the following:

(a) Maximum storage life of uncatalyzed resin and catalyzed resin at various temperatures.

(b) Recommended catalyst types, and catalyst percentages.

(c) Minimum and maximum gel time and peak exotherm of catalyzed resin.

(d) Recommended curing conditions for cast sheets from which specimens conforming to the specification can be prepared.

3.6 Certification - The manufacturer shall certify that, (a) the material is based on a reacted bisphenol -A derivative, (b) all material in each lot was produced under the same manufacturing procedure, (c) the chemical composition of the material submitted is the same as a previous lot which was subjected to chemical resistance tests (when applicable). The certificate shall be signed by a responsible agent of the manufacturer, and shall be accompanied by evidence of the agent's authority to bind his principal.

3.7 Workmanship - The material shall be free from visible contaminants and shall contain no lumps or show other signs of partial gelation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot - A lot shall consist of a quantity of resin of uniform chemical composition, produced under like conditions in one unchanged process by one manufacturer, and presented for acceptance at one time.

4.3 Inspection - Inspection for acceptance of each lot shall consist of determining conformance to the requirements of Section 3, as applicable, and the examination specified in 4.5.

4.4 Sampling

4.4.1 Sampling for examination of filled containers - A sample of filled containers shall be selected from each lot in accordance with Military Standard MIL-STD-105 at Inspection Level I to verify compliance with all stipulations of this specification regarding inspection requirements not involving tests (see Table II).

4.4.2 Sampling for tests - Samples for testing shall be selected at random from each lot. The sample shall be selected in accordance with Military Standard MIL-STD-105 at Inspection Level S-1 and shall be of sufficient size to prepare all the required specimens.

Table II - Examination procedures and standards[1/]

AQL percent	Classification of defect	Defect	Method of inspection
1.0	Major 103	Contaminated resin	Visual
	Major 102	Non-uniform resin	Visual
1.0	Major 103	Improper weight	Approved scale [2/]
4.0	Minor 201	Lack of, or improper instructions (see 3.5)	Visual
	Minor 202	Wrong size container	Visual
	Minor 203	Container improperly closed	Visual
	Minor 204	Marking misleading or unidentifiable	Visual

1/ Unless otherwise specified by the procuring activity, Standard MIL-STD-105 shall be used as a basis.

2/ Approved by the procuring activity.

4.5 Examination

4.5.1 Visual - Sample units of resin shall be examined for the defects and at the acceptable quality levels shown in Table II.

4.5.2 Packaging, packing, and marking - Examination shall be for the defects and at the acceptable quality levels shown in Table II.

4.6 Test procedures

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4.6.1 Test conditions - Unless otherwise specified, all tests shall be performed at 23° ± 1°C (73.4° ± 1.8°F). Moreover, immediately prior to testing, all specimens shall be conditioned at 23° ± 1°C, until this temperature is uniform throughout the specimen.

4.6.2 Preparation of specimens - Unless otherwise specified by the procuring activity, the specimens cited in 4.6.3 thru 4.6.6 shall be cut from 1/8 inch thick cast sheets prepared in accordance with the manufacturer's instructions (see 3.5). The resulting sheet shall be free of air bubbles and shall be representative of the best quality workmanship.

4.6.3 Flexural strength and modulus - Five specimens shall be tested in accordance with Method 1031 of Fed. Test Method Std. No. 406.

4.6.4 Tensile strength and elongation - Five specimens shall be tested in accordance with Method 1011 of Fed. Test Method Std. No. 406.

4.6.5 Deflection temperature under load - Three specimens shall be tested in accordance with Method 2011 of Fed. Test Method Std. No. 406.

4.6.6 Water absorption - Three specimens shall be tested in accordance with Procedure A of Method 7031 of Fed. Test Method Std. No. 406.

4.7 Rejection

4.7.1 Examination - Any non-conforming material specimen or container in each sample, when examined in accordance with 4.5, shall be rejected. Also, if the number of non-conforming specimens or containers exceeds the acceptance number specified for that sample, the entire lot shall be rejected.

4.7.2 Tests - A lot shall be rejected for failure to comply with any of the test requirements when tested in accordance with 4.6.

5. PREPARATION FOR DELIVERY

5.1 Application - The requirements of Section 5 apply only to purchase by or direct shipment to the Government.

5.2 Packaging - Unless otherwise specified, the resin shall be packaged in metal containers conforming with Specification PPP-C-96 and PPP-D-760.

5.3 Packing - Unless otherwise specified, the resin shall be packed in containers not to exceed 200 pounds gross weight.

5.3.1 Level A - Unless otherwise specified, packing shall be in accordance with either (a) PPP-B-601 and NN-P-515 or, (b) PPP-B-621.

5.3.2 Level C - Packing shall be in accordance with commercial practice adequate to insure acceptance and safe delivery by the carrier for the mode of transportation employed.

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5.4 Marking - In addition to any special marking specified by the procuring activity (see paragraph 6.2) all containers shall be marked in accordance with MIL-STD-129. Every container shall also be conspicuously marked: STORE IN COOL PLACE. In addition, every container shall have the following information marked on its exterior:

(a) Date packaged (Month and year)
(b) Lot number.

5.5 Instruction sheet - Unless otherwise specified, every other container shall have a copy of the manufacturers' instruction sheet meeting the requirements specified in 3.5.

6. NOTES

6.1 Intended use - This resin is intended for use in the fabrication of fibrous glass reinforced structures which are exposed to corrosive aqueous solutions.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Chemical resistance desired and method of determining conformance (see 3.3).
(c) Applicable level of packing (see 5.3).
(d) Special marking desired, if any (see 5.4).

6.3 Resins conforming to this specification are claimed to offer maximum corrosion resistance to a wide range of industrial salts, acids, alkalies, and organic compounds. However, it is recommended that where resistance to any particular corrosive environment is desired, appropriate tests be specified to determine the suitability of the material in the application. It is suggested that where the tests consist of determination of mechanical property retention after immersion in a corrosive solution, a minimum of six months immersion be specified.

6.4 Typical flexural strength retention, for a material conforming to this specification, after 12 months immersion at 210°F when tested at room temperature, are as follows:

Solution	Percent retention
5% nitric acid	23%
25% sulfuric acid	71%
15% hydrochloric acid	56%
25% acetic acid	41%
5% electrasol	82%
5% sodium hydroxide	80%
5.25% sodium hypochlorite	74%
distilled water	85%

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